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Knowledge on adolescent and reproductive health among secondary school students in Khulna City, Bangladesh

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Abstract
The main objective of this study was to investigate the depth and breadth of knowledge and ideas about adolescent health among secondary-level students. A concurrent mixed method approach, both quantitative and qualitative methods, was followed to carry out the study. Students enrolled in tenth grade under the Directorate of Secondary and Higher Education (DSHE) in Khulna City, Bangladesh, were chosen as the respondents of the study. *Kotwali* and *Khanjahan Ali Thana* of Khulna City were rationally chosen as the areas of the study. A total of 40 schools were randomly chosen, and 400 surveys, 10 surveys from each school, ideally 05 boys and 05 girls, were carried out, including 10 in-depth case studies. Findings indicated that, on average, more than half of the adolescents have knowledge gaps about adolescents and reproductive health, for example, misconceptions about the adolescent period and age range of puberty, vague understanding of the first sign(s) of physical appearances, and face critical emotional and social challenges. Both qualitative and quantitative data further showed that they hardly get sufficient cooperation and support from family and social environments in dealing with their challenges, display risky health behaviour, and make major decisions based on their self-discretionary assumptions to deal with crises. The findings of this study would provide important guidelines to parents and guardians, local and district and regional and national level GOs and NGOs health professionals who work with adolescent and reproductive health services, policymakers, academics, and researchers to create an adolescent-friendly environment.

Keywords Knowledge · Adolescent and reproductive health · Secondary School · Students · Bangladesh

1 Introduction

Adolescence, the age of transition from childhood to adulthood, is a challenging period of human life. It is a period of life during which individuals reach sexual maturity [1, 2]. This period is marked by physical growth, sexual maturation, and psychosocial achievement that generally begins between ages 8 and 14 [3]. It is also described as a stage when significant physiological and anatomical changes take place, resulting in reproductive maturity [4, 5]. In adolescence, noteworthy physical and mental changes occur in a very short period. This is why this period of human life is considered a sensitive period. However, there are more than one billion adolescents worldwide [1]. According to the estimates of the World Health Organization (WHO), there is one adolescent aged between 10 and 19 years in every five people in the world [6, 7]. Around one-quarter of the total population of Bangladesh belongs to adolescent categories, where 12% of the

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adolescents belong to the age group of 10–14 years, and other around 11% of them are in the age group of 15–19 years [8]. This statistical representation of the adolescent group reiterates the importance of adolescents and their health.

Adolescents are one of the most vulnerable groups who suffer from numerous health complications due to their inadequate knowledge about adolescent and reproductive health [7–9]. They frequently make reproductive health decisions based on their own discretion and the availability of choices, which make them vulnerable to unsafe reproductive health behaviour and inappropriate choices [1]. These discretionary choices may have substantial detrimental effects on their future reproductive health [1]. Inadequate knowledge about the consequences of unprotected premarital sex among adolescent females, for example, predisposes them to unwanted pregnancies, unsafe abortion, and sexually transmitted infections [1, 10]. In Asia and Africa, a significant proportion of youth often get infected with sexually transmitted diseases, including HIV/AIDS [11–15].

In Bangladesh, reproductive health interventions for adolescents have not been widely introduced [16]. Discussing adolescent reproductive health issues is often considered as social taboo [8, 17]. Parents and others in society sometimes prefer to skip talking with teenagers about adolescent and reproductive health [11]. Insignificant support for physical, psychological, as well as social development, including access to information and services, affect healthy transition into adulthood [16]. Consequently, there are obstacles for adolescents to receive appropriate knowledge about adolescent and reproductive health, and teenagers in Bangladesh usually have ambiguous knowledge on adolescent and reproductive health.

Social customs and cultural and religious differences create polemics on adolescent and reproductive health education [11] and could be a barrier in communication that hinders them from opportunities to acquire appropriate adolescent and reproductive health knowledge. Many international organizations, e.g., the International Council for Population Development (ICPD) held in Cairo in 1994 [18] and at the Fourth UN Conference on Women held in Beijing in 1995 [19], etc., have mentioned the significance of attending the reproductive health of adolescent males along with that of females [1]. Considering the significance of the topic and the socio-cultural perspective of Bangladesh, this study aimed at understanding the depth and breadth of teenagers' knowledge on adolescent and reproductive health.

2 Literature review

Adolescence is brief but a sensitive and significant period of human life [11]. It marks the transition from childhood to adulthood, often characterized by biological changes, psychosocial development, and sexual exploration [1, 4]. It is stage of life with specific health and developmental needs and rights, where individuals also develop knowledge and skills to manage emotions, build relationships and acquire essential attributes [20]. During puberty, the sign of physical appearance includes somatic growth, primary sexual organs development like gonads and genitals, and the change of secondary sexual characteristics like breasts and pubic hair [3]. The physical manifestations of puberty contribute to two processes: *gonadarche,* which involves the development of ovary or testes, and *adrenarche,* which involves the adrenal gland [3]. It is a time when youths discover how to navigate new, often compelling, social challenges and adjust to the myriads of physical, cognitive, and emotional changes [21–23].

Adolescent is marked as a fastest-growing segment of population than other groups to represent in health literacy interventions [24, 25]. On average, girls enter puberty earlier than boys [3]. When comparing the growth spurts that occur puberty, girls typically experience this development around the mean age 11–12 years, while boys do so at the mean age of 14 years [26]. In girls, the most dramatic sign of sexual maturity is the onset of menstruation, which usually occurs at an average age of 12.8 years, but after approximately one to two years, the menstrual cycles become ovulatory and more regular [3, 27]. The first visible sign of sexual maturation in girls is the breast bud's appearance, which generally becomes visible around age 10 or 11 [3]. On the other hand, for boys, an increase in testicular size occurs at 9.5–13.5 years of age, followed by pubic hair growth [3, 28]. The penis lengthens and widens but takes several years to become full-size [3]. Generally, at ages 13.5–14 years, sperm production coincides with testicular and penile growth occurring [3].

Teenagers' physical appearances are the signal of psychological and social changes that mark this period as a critical stage in becoming an adult [26]. They are made particularly sensitive to their social environments by the impacts of puberty on the brain [21, 29, 30]. They go through a period of social reorienting where the opinions of their peers become more important than their parents [29, 31, 32]. Peer evaluations affect their feelings of social or personal worth, and peer rejection indicates their unworthiness [29, 33]. They get new metacognitive and executive skills that allow them to express their thoughts, feelings, and actions deliberately to reason [34, 35]. They can consciously reflect on their experiences, draw conclusions, and create strategies for navigating complex situations [34].

In Bangladesh, discussion about adolescent and reproductive health is still a cultural taboo [8, 11, 36]. People do not talk to teenagers about adolescent sexual and reproductive health [11]. Parents do not feel comfortable communicating on reproductive health issues like menstruation, unwanted pregnancy, ejaculation, etc., with their adolescents [8, 11, 36, 37]. In addition, schools provide minimal information, which has severe health risk implications for teenagers [8, 11, 36]. Teachers usually try to avoid lecturing on adolescents and reproductive health, considering cultural perspectives [11]. When teenagers barely have any other sources to get adolescent health info, they mostly depend on peer information and self-discretionary assumptions [8, 11]. On the other hand, most adolescents have no idea about menstruation or wet dreams until they experience those [37]. Almost half of the teenagers could not appropriately identify a single symptom of Sexually Transmitted Diseases (STDs) or a mode of transmission of STDs [38, 39]. This misapprehension may stem from the sensitive nature of issues related to sexuality and reproductive health [12]. In such cases, they may be at risk of STDs for displaying sexual behaviours and developmental characteristics that place them at risks [4].

Generally, adolescence is a healthy period of human life, but many adolescents are less informed and less experienced [7]. Unprotected premarital sex among adolescents, unwanted pregnancies, unsafe abortion and its complications, and sexually transmitted infections, are the consequences of the lack of knowledge [1, 10, 11]. The knowledge about sexual and reproductive health and access to adolescent and reproductive health services are significant for adolescents' physical and psychosocial well-being [1, 11]. However, there is a significant gap in knowledge about the prevalence of knowledge on reproductive health among adolescents, the challenges they face, and their coping strategies to face the challenges. To this end, this study was conducted to understand the knowledge base of adolescents and reproductive health among secondary school students in Khulna City.

3 Research objective and methods

3.1 Objective of the study

The overall objective of the study was to analyse the depth and breadth of knowledge on adolescent and reproductive health among secondary school children of Khulna City, Bangladesh. The specific objectives of the study were to understand the level of knowledge of adolescence about adolescent period, and their physical and emotional appearances, and to picture the challenges teenagers face due to their lack of knowledge.

3.2 Research approach and methods

In this study, a concurrent mixed-method approach, both quantitative and qualitative methods, was applied. The mixed-method approach was appropriate, considering the complex phenomena and getting a better understanding and comprehension [11, 40–43]. The fundamental rationale of choosing a mixed-method approach was to combine the strengths of both qualitative and quantitative methods [42]. The researchers got an opportunity to verify this investigation from different ontological and epistemological points of view through this approach [42–44]. In this study, surveys as a quantitative approach and in-depth case studies as a qualitative approach were applied.

3.3 Study area and location

To carry out this study, as the third divisional and major city of Bangladesh, Khulna was selected as the study area [8, 11, 45–48] and many others also carried out their study on adolescence in Khulna District. In Khulna, the total population of 10–14 years is 72,370, with a male population of 37,131 and a female population of 35,239 [49]. Furthermore, there are 46,207 children who belong to the age category of 15–19 years, where the number of male is 25,228 and the number of female is 20,979 [49]. The Directorate of Secondary and Higher Education (DSHE) has two *Thana* in the city. One is *Kotwali Thana* and the other is *Khanjahan Ali Thana*. In the study, the *Kotwali Thana* was called 'K1', and the *Khanjahan Ali Thana* area was called 'K2'. There are one hundred secondary schools in both areas, with 59 school located in *Kotwali Thana*, and 41 schools in *Khanjahan Ali Thana*.

3.4 Sampling and respondents

In the quantitative section, the cluster sampling method was adopted in this study. Among the hundred schools under DSHE of Khulna zone, *Kotwali Thana* was marked as 'K1', and *Khanjahan Ali Thana* was marked as 'K2' in the very first step. In the second step, all the schools of *Kotwali Thana* = k1 and *Khanjahan Ali Thana* = K2 have been marked. In the third step, twenty schools were randomly selected from fifty-nine schools of the *Kotwali Thana* = k1 area, and a further twenty schools were chosen from forty-one schools of the *Khan Jahan Ali Thana* = K2 area. In the final step, 10 students of tenth grade were randomly selected from each school. In this way, two hundred boys and two hundred girls from each *Thana* between the age categories of 14 to 16 were selected for this study.

In the qualitative section, a total of 10 in-depth case studies (ICSs) were carried out where the ratio of boys and girls was equal, ideally five for each. In collecting quantitative data, some respondents were primarily marked for ICSs by observing their response quality, confidence, and flexibility with the interviewee. Among those, a total of 10 participants were approached and appointed as respondents for the qualitative section who volunteered to participate willingly. This is to note that for ICSs, not more than one respondent was selected from each school, with six respondents from *Kotwali Thana* = K1 area and four respondents from *Khanjahan Ali Thana* = K2 area.

3.5 Data collection methods and instruments

For the quantitative section, the survey method was applied, and data were collected through a face-to-face interview method using a semi-structured interview schedule. In the beginning, a pre-test was conducted with six students from two schools to identify and correct errors, evaluate question format and sequence, check the length and timing, identify and mitigate bias, and other technical issues. After the pre-test, the questionnaire was revised and edited to conduct surveys in schools. Qualitative data were collected using separate guidelines for boys and girls.

3.6 Data analysis techniques

The quantitative data collected through a semi-structured questionnaire were analysed descriptively using the SPSS (version 26.0) and presented in a tabular and graphical format. Qualitative data collected through ICSs were carefully written and compiled immediately after returning from the field from the recorded speeches. Data and information collected were stored in separate files with code numbers. Thematic approach was used to analyse qualitative data. Finally, collected data through surveys and in-depth case studies were concurrently analysed and presented using simple descriptive methods.

3.7 Ethical issues

There are no institutional review boards (IRBs) in Bangladesh that can approve this study. While it is really a concern that social science studies might go unchecked without ethical boards' approval in Bangladesh, scholars usually need to depend on source books' codes to remain ethical and maintain the standard. For example, in carrying out this study, the authors followed the ethical guidelines provided by Miles and Huberman [50]. Many research scholars followed this guideline in carrying out their studies in Bangladesh (see: [11, 40, 44, 51, 52]). However, to collect data from the field, approval from the head teacher at each school has been taken in the first stage. Upon the initial consent from the head of the schools, the research team went to each school to provide a comprehensive explanation on the subject and objectives of the study in helping students to make an informed decision. Students were instructed to talk to their parents and come back with their final decision to be or not to be a part of the study. Subject to parental consent, the majority of the students expressed their interest in being a part of the study. Students who showed reluctance to be part of the study were excluded from the list before finalizing it and making the random selection. After random selection and before the data collection process, an informed consent was taken subject to providing sufficient information about the study details. It is to be noted that an informed consent was also obtained from the parents/legally authorized representatives of subjects who were under 16. However, all participants were assured that the information collected from them would only be used for this study. All the names used in this study

Table 1 Sample size and respondents Developed by authors

a: Respondents sample for quantitative section

Gender	Kotwali Thana = K1	Khanjahan Ali Thana = K2	Total
Boys	100	100	200
Girls	100	100	200
Total	200	200	400

b: Case study list

Case Number	Name	Gender	Age	School Name	Thana Area
First	Afnan	Male	14	Khulna Zilla School	Kotwali
Second	Moutushe	Female	14	Govt. Coronation Secondary Girls' School	Kotwali
Third	Tamim	Male	14	ST Joseph's High School	Kotwali
Fourth	Adhora	Female	14	Khulna Collegiate Girls School	Kotwali
Fifth	Talha	Male	15	Bangabasi High School	Khanjahan Ali
Sixth	Sorna	Female	14	Crescent Secondary School	Khanjahan Ali
Seventh	Raisa	Female	14	Port Secondary School	Khanjahan Ali
Eighth	Ratul	Male	14	Rotary School Khulna	Kotwali
Ninth	Atifa	Female	15	Khulna Govt. Girls School	Kotwali
Tenth	Nadim	Male	14	Govt. Laboratory High School	Khanjahan Ali

are pseudonyms, considering confidentiality. The research team responded to if any questions or confusions arose during or after the data collection process (Table 1).

4 Result and findings

4.1 Understanding about adolescent health

Findings revealed that regarding the concept of adolescence, most participants responded that it is both a physical and emotional appearance in the human body (69%), while 22% of the respondents termed it as a transition period from childhood to adulthood (*See* Table 2a). Aligning with survey findings, participants of ICSs also defined the age of adolescence as a period of transition between childhood and adulthood. For example, one respondent (fourth case study) stated-

> Adolescence is the period between the transitions from childhood to adulthood. (Adhora, 14 years, Khulna Collegiate Girls School, Kotwali Thana, Khulna)

Another respondent (sixth case study) explained that,

> Adolescence is a time period where no one is a child and also no one is an adult. (Sorna, 14 years, Crescent Secondary School, Khanjahan Ali Thana, Khulna)

More than half of the participants barely have a specific understanding of the age range of puberty and the benchmark age of adolescence. About the adolescent age range, nearly one-third of the participants (31.0%) marked 10–19 appropriately as the benchmark age for the adolescent period *(see* Table 2b*)*, and sharply half of the participants (50%) rightly marked 12–13 years as the age to start puberty (*see* Table 2c*)*. About the adolescent period, one respondent (third case study) expressed that,

> Adolescent symptoms are observed between 10 and 13 years of age, and its duration between 12 to 19 years. (Tamim, 14 years, ST Joseph's High School, Kotwali Thana, Khulna)

Another respondent (ninth case study) included that,

> In adolescence, girls enter earlier than boys. Girls may enter at age 11-12 years and boys may enter at age 12-13 years. (Atifa, 15 years, Khulna Govt. Girls School, Kotwali Thana, Khulna)

Table 2 Adolescence

Response	K1 Boys (%)	K2 Boys (%)	K1 Girls (%)	K2 Girls (%)	Overall (%)
a. Concept about adolescence					
Transition period from childhood to adulthood	23.0	25.0	28.0	10.0	21.5
Some physical appearances occur in the human body	10.5	7.5	4.0	14.0	9.0
Both physical and emotional appearances occur in the human body	66.6	67.5	68.0	74.0	69.0
Don't Know	00	00	00	2.0	0.5
Total	100.0	100.0	100.0	100.0	100.0
b. Adolescent period range					
9–16 age	00	2.0	00	00	0.5
10–15 age	14.0	10.0	42.0	28.0	23.5
10–16 age	37.5	44.5	36.0	62.0	45.0
10–19 age	48.5	43.5	22.0	10.0	31.0
Total	100.0	100.0	100.0	100.0	100.0
c. Puberty Started					
10–11 age	35.0	27.7	51.5	36.0	37.5
12–13 age	45.0	48.5	42.5	64.0	50.0
14–15 age	20.0	10.0	6.0	0	9.0
15–16 age	0	14.0	0	0	3.5
Total	100.0	100.0	100.0	100.0	100.0

Source: Field Survey

Boys of both groups showed similar and reasonable understanding of the changes in physical appearance. For example, regarding the changes in boys' physical appearances, voice cracks and gets deeper (13%), height and weight increase (12%), and underarm and facial hair growth (12%) have been selected most, followed by the growth of the penis, pubic hairs, and wet dreams (see Table 3a). Though boys are supposed to mark all the options to show their comprehensive knowledge on physical appearance issues, only on average, 6% of respondents selected all the appearances. On the other hand, regarding the changes in physical appearances among girls, findings revealed that both groups have a very close understanding. The most common factors the girls' group chose were the occurrence of menstrual period (15%), enlargement of breasts (14%), and pubic hair growth (14%) (see Table 3b). Similar to boy groups, only 5% of the total girls marked all the options, meaning that they lack comprehensive knowledge regarding adolescent physical appearances (Fig. 1).

About the emotional appearances (see Table 3c), all the options were correct, and put to see how many participants tick all of them. It was found that participants have a lack of understanding as they did not choose all the available options but rather chose one or a few options. Participants selected gaining sexual feelings (14%), feeling conscious about self (13%), and mood swings (13%) as the major signs of emotional appearances. However, there were differences in understanding about the emotional appearances between boys and girls, for instance, while boys mostly chose sexual feelings, conflicting thoughts, and looking for an identity, girls generally picked options like peer pressure, getting sexual feelings, and feeling uncertain. Furthermore, the sources of gaining credible knowledge are too scarce as they are frequently denied by their parents and relevant others who could be resources to them. For instance, regarding emotional appearance, one girl respondent (seventh case study) confessed that,

> Sometimes I am uncertain about myself for many reasons. When all my friends had been started menstruating but not mine. So, I went to my mother to share about it. But she became angry with me and scolded me. (Raisa, 14 years, Port Secondary School, Khanjahan Ali Thana, Khulna).

A boy respondent (first case study) also accused that,

> All my friends have become much taller than me. They also have beards on their faces. However, I still cannot see the beard on my face. That's why I am worried about myself. (Afnan, 14 years, Khulna Zilla School, Kotwali Thana, Khulna)

Boys marked that voice cracks (24%) and wet dreams (24%) as their first sign of the physical appearance, while girls marked menstrual period (50%) as their first sing of physical appearance (see Fig. 2). Findings of the ICSs also

Table 3 Appearances in Adolescence (Multiple Choice)

Appearance	K1 Boys (%)	K2 Boys (%)	K1 Girls (%)	K2 Girls (%)	Overall (%)
a. Boys Physical Appearances					
Height and weight increase	11.7	12.0	12.9	13.0	12.4
Muscles develop	8.1	8.7	7.2	6.8	7.7
Voice cracks and gets deeper	11.7	11.8	12.2	14.7	12.6
Underarm and facial hair grow	11.3	11.9	12.2	13.6	12.2
Body Odor and acne	9.2	8.7	8.7	9.1	8.9
Penis grows longer and thicker	11.7	11.5	11.9	11.1	11.5
Pubic hair grows	10.6	11.7	10.3	11.8	11.1
Testicles grow larger	7.4	6.7	6.1	3.2	5.9
Wet dreams occur	11.7	11.9	11.5	11.5	11.7
Too much sweating	6.6	5.1	6.0	5.0	5.7
Not Answered	00	0.0	1.0	0.2	0.3
b. Girls Physical Appearances					
Body fat increases	6.8	5.6	4.5	3.3	5.1
Breasts begin to enlarge	13.7	14.3	14.1	14.2	14.1
Pubic hair grows	13.2	13.0	14.1	14.9	13.8
Height and weight increase	12.7	12.8	11.9	15.2	13.1
First menstrual period occurs	14.1	15.0	15.8	16.2	15.3
Hips widen	8.3	7.0	7.4	6.0	7.1
Underarm hair grows	10.2	10.1	13.5	16.2	12.5
Skin and hair become more oily	7.7	9.4	6.8	6.1	7.5
Pimples may appear	10.9	11.6	11.9	7.9	10.6
Not Answered	2.4	1.2	00	00	0.9
c. Boys and Girls Emotional Appearances					
Feeling overly sensitive	8.7	10.0	10.0	8.5	9.3
Looking for an identity	13.9	11.7	12.8	11.5	12.5
Feeling uncertain	10.1	12.5	13.1	14.9	12.7
Peer pressure	9.6	11.1	15.6	12.5	12.2
Conflicting thoughts	15.4	12.5	11.4	14.5	13.4
Mood swings	13.9	12.5	12.1	12.8	12.8
Feeling conscious about self	13.0	14.0	11.8	12.8	12.9
Getting sexual feelings	15.4	15.7	13.2	12.5	14.2

Source: Field Survey

reflected the same understanding as found in surveys, for instance, wet dream/penile ejaculation and menstruation, etc., were narrated as the first signs of adolescence. One boy respondent (fifth case study) described:

> In my experience, boys enter puberty when they have a wet dream or penile ejaculation, so the wet dream is their first physical appearance during puberty. (Talha, 15 years, Bangabasi High School, Khanjahan Thana, khulna)

Another respondent (second case study) acknowledged that,

> I guess that boys have already entered puberty when their beards twist. (Moutushe, 14 years, Govt. Coronation Secondary Girls' School, Kotwali Thana, Khulna)

One girl respondent (sixth case study) revealed that,

> Girls enter puberty when the first menstrual period occurs, so the first menstrual period is the first appearance of puberty in girls. (Sorna, 14 years, Crescent Secondary School, Khanjahan Ali Thana, Khulna)

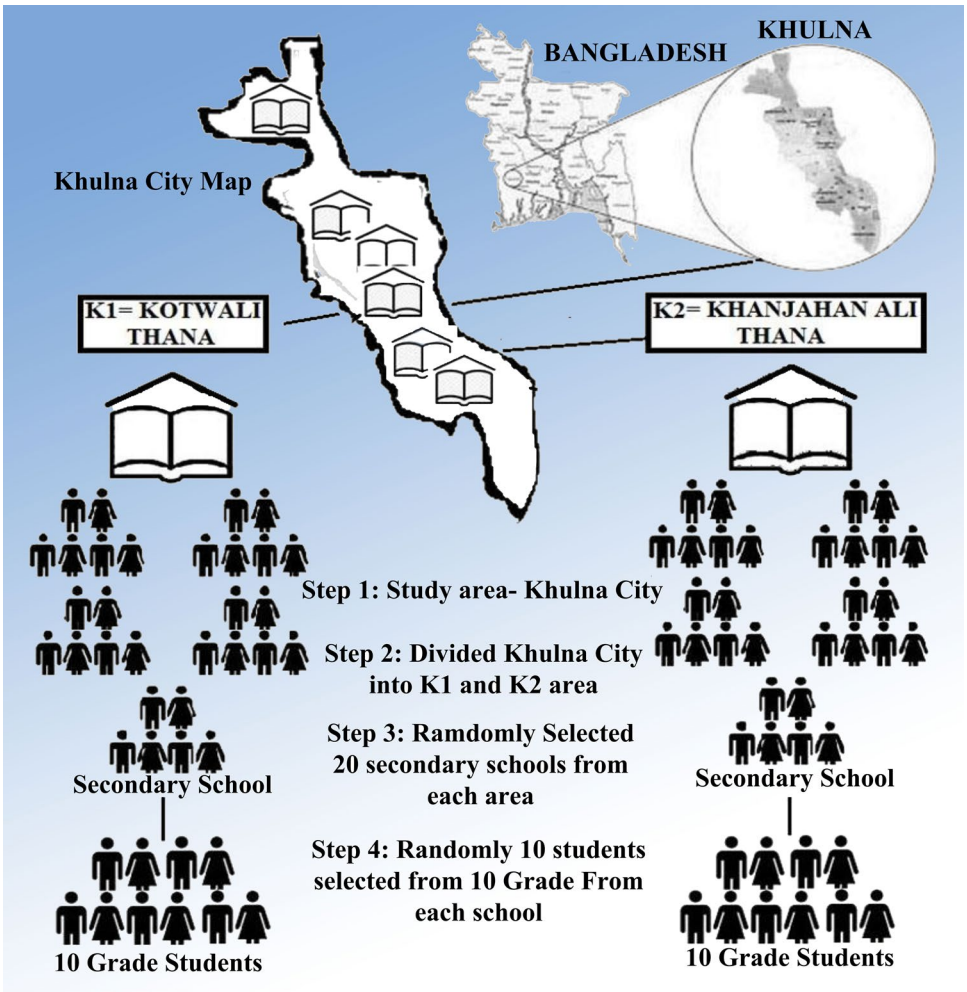


Fig. 1 Cluster sampling method Developed by authors

5 Conception on significant appearances in adolescence

Eight out of ten (80%) of the adolescents have a clear conception of the place where public hair grows, ideally by marking underarms and genital areas. The rest of the participants picked either one of them or said that it might grow in many places of the human body (12%) or responded with 'do not know' (2%) (see Table 4a). However, on average, both boys and girls were found to have almost the same level of understanding about pubic hair. One of the respondents (seventh case study) exposed that,

> Not sure, but I think pubic hair is one kind of hair that can be seen under the armpits and around the genitals. Because, in my experience, hair starts to grow in these places as soon as puberty starts. (Raisa, 14 years, Port Secondary School, Khanjahan Ali Thana, Khulna)

During adolescence, among the physical appearances, the most sensitive appearance that occurs in boys is penile ejaculation or wet dreams. Nearly half and a quarter of the participants remarked 'wet dream' as a natural condition that something happens in dreams and the fluid containing sperm comes out from the pennies (73%) (see Table 4b) and other 14% of the participants marked it as a sexual situation that happens asleep at night. Here, only 9% of the respondents remarked on the appropriate ideas. One boy respondent (tenth case study) apprised that,

> While asleep, the boys ejaculate alone as a result of seeing something in their dreams. This type of event is called wet dreams. (Nadim, 14 years, Govt. Laboratory High School, Khanjahan Ali Thana, Khulna)

Another girl respondent (seventh case study) confessed that,

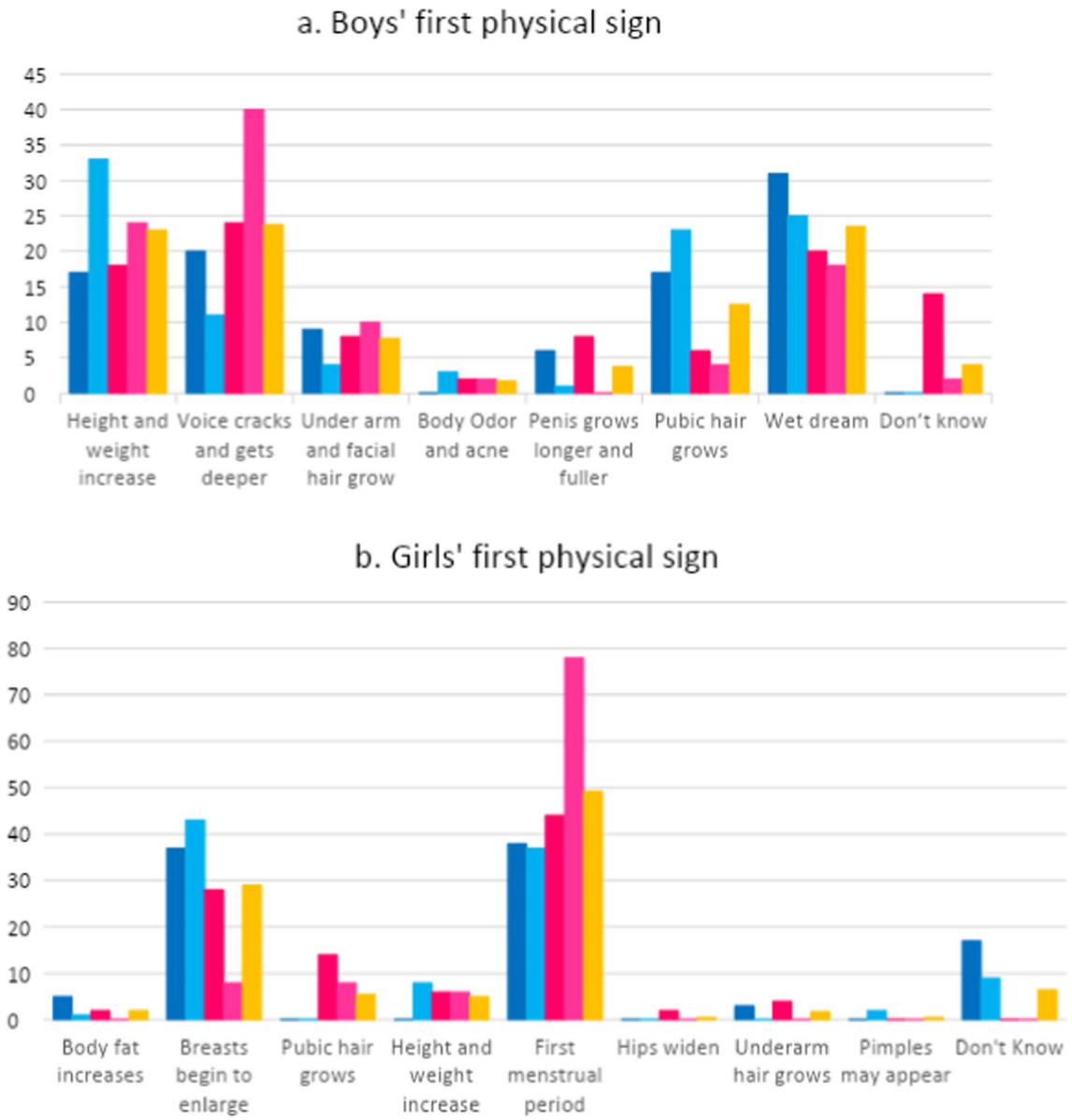


Fig. 2 First sign of physical appearance in puberty. Source: Field Survey

As far as I've heard, the wet dream is a physical issue for boys. Boys ejaculate from their penis during sleep. I don't know more than that. (Raisa, 14 years, Port Secondary School, Khanjahan Ali Thana, Khulna).

The girls' menstrual period is also a sensitive appearance during adolescence period. Two quarter of the participants (66%) stated that it is natural to occur for the girls that blood come out from their vagina in a month. In this case, the ratio of girls was higher than boys to pick the option. However, only 14% of the participants were able to pick the right answer, ideally 'a cycle of time that includes bleeding from the female vagina' (see Table 4c). To this end, it is to be noted that girl's ratio was comparatively lower than boys, and only 3% of the girls from K2 groups was able to pick the appropriate option. This reflects the substantial knowledge gaps among adolescents about menstruation period. About the girls' menstrual period, one boy respondent (tenth case study) revealed that,

Menstruation is a physical matter in girls that causes blood to come out of their vaginas on certain days of the month. (Nadim, 14 years, Govt. Laboratory High School, Khanjahan Ali Thana, Khulna)

Another boy respondent (first case study) confessed that,

Table 4 About pubic hair, wet dream, menstrual period and emotional appearance

Response	K1 Boys (%)	K2 Boys (%)	K1 Girls (%)	K2 Girls (%)	Overall (%)
a. Pubic Hair					
It is one kind of body hair, which is growing many places in human body	12.0	16.0	10.0	10.0	12.0
It is hair that is growing in underarm	00	00	2.0	00	0.5
It is body hair found in the genital area	7.0	5.0	10.0	2.0	6.0
It is both underarms and genital hair	79.0	77.0	78.0	86.0	80.0
Don't know	2	2.0	00	2.0	1.5
Total	100.0	100.0	100.0	100.0	100.0
b. Wet Dream					
It is one kind of urine that discharges asleep	4.0	2.0	1.0	3.0	2.5
It is the sexual situation that happens asleep at night	16.0	14.0	13.0	11.0	13.5
It is a normal condition that discharged the fluid containing sperm from the penis while a boy's asleep	15.0	13.0	4.0	2.0	8.5
It is natural condition that something is happening in dreams and the fluid containing sperm come out from the pennies	65.0	71.0	75.0	79.0	72.5
Don't know	00	00	7.0	5.0	3.0
Total	100.0	100.0	100.0	100.0	100.0
c. Menstrual Period					
It is a period that girls become sick	12.0	8.0	2.0	6.0	7.0
It is the time for every girl that blood comes out from their vagina in the certain time	8.0	10.0	2.0	00	5.0
It is a cycle of time that includes bleeding from the female vagina	22.0	12.0	19.0	3.0	14.0
It is natural to occur for the girls that blood come out from their vagina in a month	45.0	51.0	77.0	91.0	66.0
Don't know	13.0	19.0	00	00	8.0
Total	100.0	100.0	100.0	100.0	100.0
d. Experiences with Emotional Appearance					
Sexual feeling	11.0	7.0	8.0	10.0	9.0
Become overly sensitive about their new physical appearance	33.0	35.0	55.0	63.0	46.50
Attraction in opposite gender	33.0	39.0	15.0	11.0	24.50
Sexual arousal	23.0	19.0	22.0	14.0	19.50
Don't Know	0.0	0.0	0.0	2.0	0.50
Total	100.0	100.0	100.0	100.0	100.0

Source: Field Survey

I don't have enough idea about the girls' menstruation. However, as far as I know, it is a kind of physical process in girls. If this is the case, the contaminated blood of the girl's body is excreted through their vagina. (Afnan, 14 years, Khulna Zilla School, Kotwali Thana, Khulna)

One girl respondent (seventh case study) stated that,

Menstruation is a physical process in girls that causes blood to come out of their vaginas after passing certain days of the month. (Raisa, 14 years, Port Secondary School, Khanjahan Thana, Khulna)

In adolescence, boys and girls experience to new emotional appearances. Nearly half of the participants marked that during adolescence, they became overly sensitive about their new physical appearance (47%). In this case, the ratio of girls was almost double that of boys. About the feelings of the opposite gender, the ratio of boys (36%) was around three times higher than girls (13%) (see Table 4d). Regarding the emotional appearance, one girl respondent (sixth case study) described that,

Honestly, when anybody looked at me from head to toe, I felt a kind of discomfort in my body, as well as a lot of shame, which I still feel. (Sorna, 14 years, Crescent Secondary School, Khanjahan Ali Sub-district, Khulna).

Another girl respondent (ninth case study) delineated that,

Suddenly, my breasts began to enlarge and quickly took on a shape. Which made me feel very uncomfortable. When someone looked at me, it seemed that he was paying more attention to my breasts. Due to this, I was very ashamed and because of this, I did not want to go in front of anyone easily. (Atifa, 15 years, Khulna Govt. Girls School, Kotwali Sub-district, Khulna).

One boy respondent (eighth case study) briefed that,

I noticed that suddenly a different kind of feeling began to work towards girls. In particular, I had a kind of attraction towards girls older than me. Due to this, I felt very ashamed to go to them. (Ratul, 14 years, Rotary School Khulna, Kotwali Thana, Khulna).

Another boy respondent (fifth case study) acknowledge that,

If I saw a romantic scene in my drama-movie, or suddenly remembered that scene, my penis would become hard, long, and straight. It made me feel very uncomfortable and ashamed. I also felt very sinful. (Talha, 15 years, Bangabasi High School, Khanjahan Ali Thana, khulna).

6 Problems faced by adolescents due to the knowledge gap

In adolescence, boys and girls face critical challenges due to their new physical and emotional appearances. The challenges include acne, facial hair, underarm hair, pubic hair, penile ejaculation or wet dream for boys, and menstrual maintenance and adjustment for girls. Around one third of the participants stated that they did not notice acne/pimples (32%). However, one out of six of the participants (18%) were scratching the acne/pimple with their own nails, and one in every seven participants (14%) were rubbing ice where it grows (see Table 5a). These reflect the wrong measures adolescents take due to their lack of knowledge. For body odor, results showed that 18% of participants have taken the best right initiative to protect themselves by using deodorant while, other 39% used body spray and 24% used perfume, a close best initiative to deal with body Odor (see Table 5b).

For underarm hair, boys mostly said to shave it (52%), while using hair removal was the most common method among girls (64%). However, one third of the boys (33%), including a portion of the girls, are used to using risky measures to deal with underarm hair, ideally the use of scissors. Furthermore, while it is a matter of hygiene, 11% of the participants said not to take any action (see Table 5c). Boys and girls responded by applying almost the same methods to deal with pubic hair. ICSs findings also align with the survey findings. Though boys get their underarm hair get removed from the barber, they take risky measures when it comes to pubic hair. Girls seem more concerned and use hair removal. However, the use of the father's razor in secret or the shameful smile of the shopkeepers make it harder for the adolescents to get proper cleaning materials and keep them hygienic. For example, one boy respondent (tenth case study) briefed that,

Table 5 For the acne/pimple, body odor, underarm hair and pubic hair

Response	K1 Boys (%)	K2 Boys (%)	K1 Girls (%)	K2 Girls (%)	Overall (%)
a. For Acne/Pimple					
Take medicine	13.0	9.0	6.0	8.0	9.0
Wash face with face wash	26.0	20.0	8.0	6.0	15.0
Rubbing ice where grows pimples	3.0	5.0	30.0	18.0	14.0
Scratching with own nail	13.0	17.0	6.0	12.0	12.0
Nothing	17.0	19.0	24.0	10.0	17.5
Use Toothpaste	2.0	00	00	00	0.5
Drink Much Water	00	00	02.0	00	0.5
Didn't notice acne/pimple	26.0	30.0	24.0	46.0	31.5
Total	100.0	100.0	100.0	100.0	100.0
b. For Body Odor					
Nothing	17.0	19.0	10.0	20.0	16.5
Use Body Spray	49.0	43.0	35.0	27.0	38.5
Use Deodorant	16.0	22.0	20.0	12.0	17.5
Use Perfume	15.0	11.0	31.0	37.0	23.5
Didn't notice Body Odor	3.0	5.0	4.0	4.0	4.0
Total	100.0	100.0	100.0	100.0	100.0
c. For Underarm Hair					
Nothing	16.0	12.0	4.0	12.0	11.0
Shave	63.0	41.0	16.0	20.0	35.0
Cut with scissor	21.0	45.0	6.0	4.0	19.0
Trimmer	00	2.0	00	00	0.5
Waxing	00	00	2.0	8.0	2.5
Use Hair removal	00	00	72.0	56.0	32.0
Total	100.0	100.0	100.0	100.0	100.0
d. For Pubic Hair					
Nothing	6.0	14.0	6.0	4.0	7.5
Shave	57.0	47.0	14.0	32.0	37.0
Cut with scissor	35.0	41.0	2.0	00	19.5
Trimmer	2.0	00	00	00	0.5
Waxing	00	00	6.0	2.0	2.0
Use hair removal	00	00	72.0	62.0	33.5
Total	100.0	100.0	100.0	100.0	100.0

Source: Field Survey

After cutting my hair, I have my underarm hair shave at the barber but my pubic hair, I shave with the razor. (Nadim, 14 years, Govt. Laboratory High School, Khanjahan Ali Thana, Khulna)

Another boy respondent (first case study) confessed that,

My underarm hair is cut by the barber, but my pubic hair is cut by myself with a scissor. Because when I try to shave with a razor, my skin is often cut. (Afnan, 14 years, Khulna Zilla School, Kotwali Thana, Khulna)

One girl respondent (ninth case study) revealed that,

For my underarm and pubic hair, I use hair removal. It is much easier and safer to use for me. (Atifa, 15 years, Khulna Govt. Girls School, Kotwali Thana, Khulna)

Another girl respondent (fourth case study) alleged that,

I shaved my underarm hair and pubic hair with a new blade on my father's shaving razor. I want to use hair removal, but when I go to the shop and ask for a hair removal, the shopkeeper gives a strange kind of smile.

Due to this, I feel ashamed and uncomfortable to buy the hair removal from the store. (Adhora, 14 years, Khulna Collegiate Girls School, Kotwali Thana, Khulna)

Male penile ejaculation or wet dreams is a very sensitive physical appearance for boys. Regarding the ejaculation or wet experiences, four out of ten boy participants (39%) hide the matter, more than one-third of the boys (35%) said to feel uncertain, and 12% said to get scared (see Fig. 3a). Had any idea before first wet dreams, more than fifty percent of the K1 boys (57%) and two thirds of the K2 boys (65%) had not any idea before (see Fig. 3b). Survey results were substantiated by the findings of ICSs. For example, about ejaculation, a boy respondent (third case study) shared that,

Before my first experience, I did not have a good idea about this. My first penile ejaculation happened in my sleep. At first, I did not understand the matter. So I got scared. Later I realized that I had ejaculated in my sleep. I was ashamed then. For a long time, I did not tell anyone about it. (Tamim, 14 years, ST Joseph's High School, Kotwali Thana, Khulna)

Another boy respondent (eighth case study) stated that,

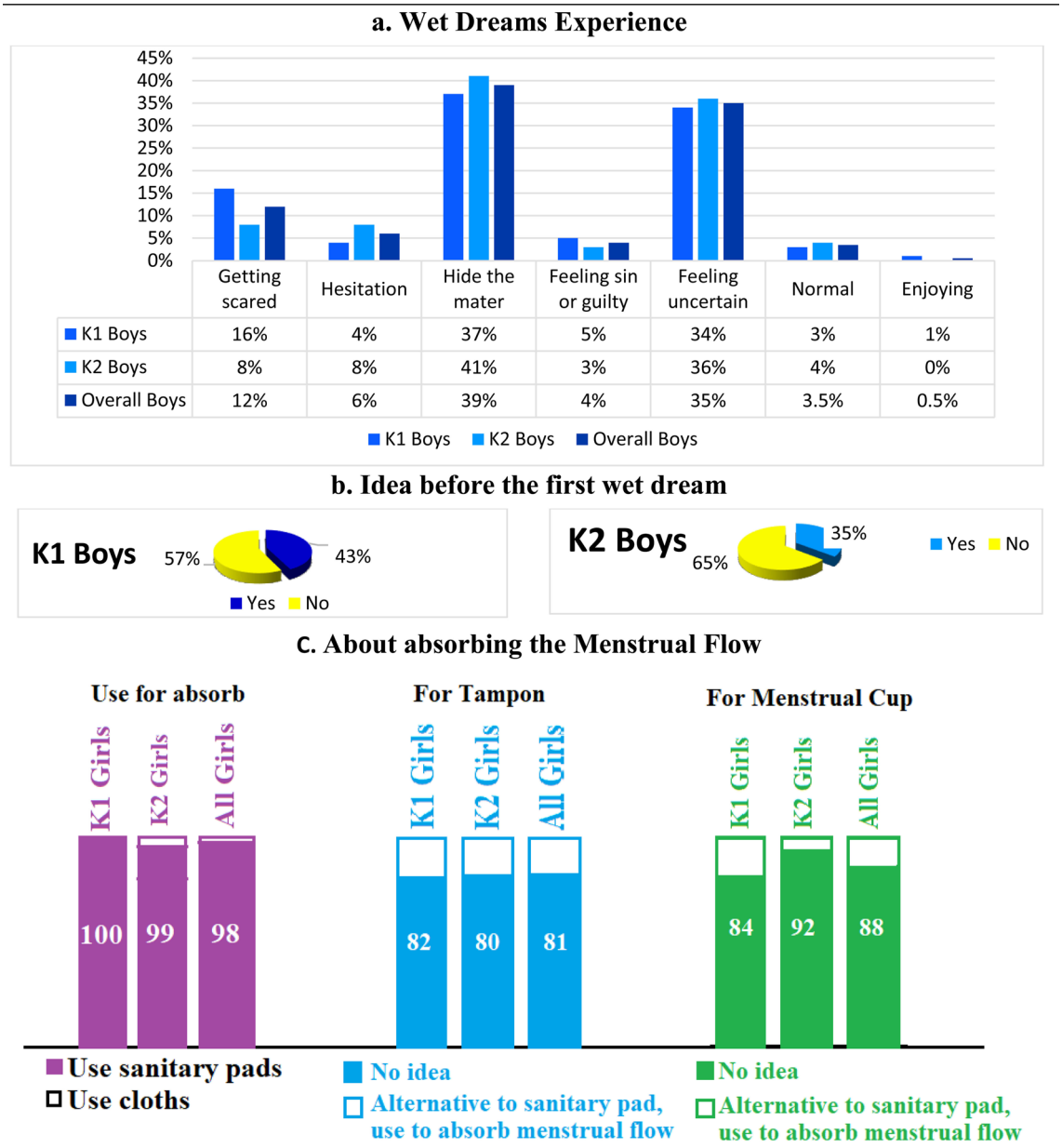


Fig. 3 About the male penile ejaculation (Wet dream) and girls menstrual Flow. Source: Field Survey

I often have wet dreams. I am very worried about it. No friend of mine has as many nightmares as I do. Many have only been there once. My friends say that I am physically very weak. So, it happens to me again and again. I can't share this with anyone but my friends. I am very worried about it. (Ratul, 14 years, Rotary School Khulna, Kotwali Thana, Khulna)

One more boy respondent (fifth case study) explained that,

After watching a romantic scene in a movie or pornography, I feel a kind of uncomfortable feeling in my penis. This feeling lasts as long as I don't masturbate. And if I don't, it will cause ejaculation in my sleep. (Talha, 15 years, Bangabasi High School, Khanjahan Ali Thana, khulna)

Menstrual flow is a very sensitive physical appearance for girls. About absorbing menstrual flow, Fig. 3c, described, 99% of the respondents used sanitary pads or napkins for absorbing menstrual flow. Only 2% of K2 girls confessed that they used clothes. About the tampon and menstrual cup, Fig. 3c, 81% of respondents acknowledged that they have not any idea about the tampon and 88% of the respondents on menstrual cup. One girl respondent (seventh case study) shared about that,

In my first menstrual period, my mother gave me a small pillow made of cotton and an old cloth inside a pillow cover. I was sitting after wearing it. I couldn't walk, I couldn't do anything. One of my aunts came to our house and saw me in this condition. She went out and bought a packet of sanitary pads for me and gave it to me. Then changed that and wore the sanitary pad. (Raisa, 14 years, Port Secondary School, Khanjahan Ali Thana, Khulna)

Another girl respondent (ninth case study) mentioned that,

I like to use sanitary pads during my menstrual period. But buying sanitary pads from a medicine store in my residential area is a complicated matter for me. There is no other woman in the family except my mother. So I can't tell anyone in the family to bring sanitary pads for me. So, my friend and I buy the sanitary pad from a great distance from our residential area. (Atifa, 15 years, Khulna Govt. Girls School, Kotwali Thana, Khulna)

One more girl respondent (fourth case study) included that,

There are men in all the medicine shops and there are a lot of people gather in those shops. Because of this, I feel very embarrassed and irritated to ask Sanitary Pad from the shopkeeper. So, I ask my mom to bring my sanitary pad. (Adhora, 14 years, Khulna Collegiate Girls School, Kotwali Thana, Khulna)

About the Tampon and Menstrual Cup, One respondents (second case study) acknowledged that,

I don't know about Tampon, but I do know about Menstrual Cup. However, I've never seen those in my eyes, and I don't know where to find those. (Moutushe, 14 years, Govt. Coronation Secondary Girls' School, Kotwali Thana, Khulna)

7 Discussion

The present study explored the prevalence and the depth and breadth of knowledge among secondary school students on adolescent health, physical and emotional appearances, and the challenges they face throughout the transition period. A concurrent mixed-method approach was used to carry out the study. Findings revealed that more than one-fifth of the teenagers (22%) meant by the notion 'adolescence' as the period between the transitions from childhood to adulthood. Numerous scholarly articles also defined adolescence as a period of transition from childhood to adulthood [4, 53, 54]. The findings aligned with Kapur [55], who explained that adolescence is a life stage between childhood and adulthood and is closely associated with the teenage years. However, around eight out of ten (79%) of the respondents explained the concept in other ways, which indicates the absence of their understanding of the conceptual definition of adolescence.

The period of adolescence is marked by the age of 10–19 years [56, 57]. This study found that approximately one-third of students marked the adolescent period accurately, stating the age range of 10–19 years (31%). The rest two-thirds of the participants wrongly marked the age range of adolescence as either 10–16 years (45%) or 10–15 years (24%). Furthermore, the study by Kapur [55] described that adolescence begins around 10–12 years, with variations among cultures and societies, and concludes between 18 to 21 years of age. It was found that sharply half of the students rightly marked that puberty

starts between the ages of 12 and 13 years, while half of the participants do not have the right understanding about the starting age of puberty. During adolescence, girls typically enter puberty around 10 or 11 years old and continue until age 16, while boys often enter around 12 years old and continue until age 16 or 17 [11, 58]. Results of ICSs revealed that girls enter puberty earlier than boys, ideally between the ages of 11–12 years and 12–13 years for boys. The findings are substantiated by studies of Kapur [55] and Caissy [59], who stated that the girls' growth spurt is around 11.5 years, and they mature two years ahead of boys.

WHO [56] explained that in boys, the first signs of pubertal change may be the tests and scrotum begin to change visibly, and in girls, the first signs of puberty may be the breast budding or light pubic hair visibility. The study by Carel and Léger [60] delineated that in tanner stage 2, the onset of puberty is marked by breast development in girls and testicular enlargement in boys. Regarding the signs of physical appearance of boys, they marked boys' voice cracks (24%), and some teenagers marked wet dream or penile ejaculation (24%). In addition, regarding the first sign of physical appearance in girls, half of teenagers mentioned the first menstrual flow (49%) while, approximately one-third of teenagers referred to enlargement of breasts (29%). The findings are substantiated by Pinyerd and Zipf's [3] study, which stated that the first visible sign in girls is the breast bud's appearance, which generally becomes visible around age 10 or 11. All these findings delineate the gaps in knowledge and misunderstandings adolescents have about the first sign of puberty.

In male penile ejaculation or wet dream, around two thirds of the teenagers (73%) have misunderstood it, and described it is a natural condition when something happens in dreams and the fluid containing sperm come out from boys' penis. However, only 9% of teenagers have appropriately termed it as a normal condition, discharging the fluid containing sperm from the penis while a boy asleep. This finding is unique to this study. On the other hand, about first wet dream or ejaculation, three quarter of the boys said to either hide the issue or feeling uncertain about what to do. The findings are justified by the study of Janssen [62], who found that a wet dream is a more obscure cultural phenomenon than an ejaculation [61]. For example, in an Indonesian study, 52% of men did not talk to anyone about wet dreams before having them. On the other hand, in a Chinese study, the author reported that 66.2% of male students have heard about wet dreams before their first experience [63]. So, from a cultural perspective, it may be a hindrance or boredom to talk about male penile ejaculation in many societies like Bangladesh.

About the girls' menstrual period, the teenagers termed it as a state when blood comes out from girls' vaginas in a month (66%). The study by Chandra et al. [8] also found that knowledge about menstruation among teenage girls was low (25.5%), medium (47.0%), and high (27.5%). Another study by Hossain et al. [12] found that knowledge about menstruation was low (38.8%), medium (53.4%), and high (7.8%) among teenage girls. This study found that girls do not get enough support from family and society during their menstrual period (fourth, seventh, and ninth case study). The findings are substantiated by studies by Chandra et al. [8] and Rashid [57]. About the menstrual flow absorbing materials, this study is potentially the first to document that teenage girl dot does not have enough ideas about tampons (81%) and menstrual cups (88%). They use Sanitary napkins/pads (99%). These findings display the teenagers' gaps of knowledge about adolescents and reproductive health.

In 2016, with assistance from UNICEF, UNFPA, WHO, and other relevant stakeholders, the Ministry of Health and Family Welfare, Bangladesh, through the General Directorate for Family Planning (GDFP) have developed and adopted the National Strategy for Adolescent Health 2017–2030 [64], aiming to ensure a healthy and productive life in a socially secured and supportive environment through easy access to quality and comprehensive information, education and services [65]. The Ministry of Education has included adolescent health issues in the school curricula. On the other hand, the Ministry of Social Welfare provides valuable support to children and adolescents, and the Ministry of Women and Children Affairs runs several projects and programs that aim to ensure the welfare of adolescents [64]. Nevertheless, the study findings revealed that teenagers still lack knowledge on adolescent and reproductive health and access to sufficient information, while almost half of the time has been elapsed since the policy was initiated. Hence, the questions arise: Why does a major portion of adolescent students still not have a broader understanding of adolescent health? What are the social, economic, and personal costs of insufficient knowledge on adolescent health for teenagers? Is the inclusion of contents in textbooks enough, or do we need to put a greater focus on family education? These are the vital questions the paper leaves for academicians, researchers, relevant officials, and policymakers.

8 Limitations and further research scope

The caution should be made that the study sampled participants from Khulna City only, which might limit the generalizability of the findings. It is to be noted that Bangladesh shares the collective culture and it is highly likely that the scenario throughout the country should be the same. Very specifically, the southern part of Bangladesh shares the same culture, and the findings should broadly be generalizable there. However, to this end, further research is recommended at a broader scale, having representative sampling from both the southern and the northern parts of the country. Furthermore, as half of adolescents have significant knowledge gaps on their health issues, further longitudinal studies can be initiated to see the long term impact of knowledge gaps on reproductive health, healthy marriage, and so on.

9 Conclusion and recommendations

This study pointed out that in average around half of the teenagers have insufficient knowledge and are less likely to gain through the conventional educational system. In addition, teenagers face many critical challenges in their new physical and emotional appearance, like being overly sensitive about their new physical appearances, girls' menstrual period, and boys' wet dreams. They feel uncertain about their life. However, they do not get sufficient support from family and society to deal with these challenges, and eventually may face critical physical or psychosocial problems. Hence, it is essential to initiate an appropriate adolescent friendly health education system and social environment to support adolescents in their sensitive life periods. Concerned with the social perspective of adolescent and reproductive health education, the study recommends-

At first, it is essential to focus on the teaching system. It should be necessary to change the conventional teaching method and find a suitable teaching method to teach teenagers about adolescent and reproductive health so that teenagers can avail themselves of true insights that will help them to grow with practical information. While it is challenging to teach adolescent and reproductive health in a country like Bangladesh, considering cultural values, hence it would be appropriate that male teachers teach boys, and female teachers teach girls separately. Moreover, teachers must also be trained as if they know how to teach such sensitive issues in culturally sensitive ways. These approaches will help to create a friendly environment for teaching and disseminating sensitive issues like adolescent health in a country like Bangladesh.

Parents should be aware of their teenagers. They must take care of their children into their teenage years. They should know how to deal with their adolescent children in their sensitive period through creating a flexible relationship so that their children can expose their physical and emotional problems to them. Considering religious and social norms and values, interventions targeting parents should have the best outcomes. This will eventually help the father become a friend of his son(s) and mother of her daughter(s) or parents of their children during the critical moments of their children. To this end, parental sensitization program needs to be initiated at a broader scale. Finally, to create an adolescent-friendly environment for greater support for teenagers regarding adolescent and reproductive health, the coalition between GOs and NGOs bears great importance. Enhanced workshops and seminars, training for teachers to teach students in a culturally sensitive way, and door-to-door campaigns targeting parents for greater awareness in the local, semi-urban, and urban areas might play a crucial role in creating a safe and friendly environment for adolescents.

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Declarations

Competing interests The authors declare no competing interests.

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